

## Multiple Choice (10 marks)

1. Which of the following language features requires that stack-based storage allocation be used rather than static allocation?
  - (a) Recursive procedures
  - (b) Arbitrary goto's
  - (c) Two-dimensional arrays
  - (d) Integer-valued functions
2. What was the first buffer overflow attack?
  - (a) Love Bug
  - (b) SQL Slammer
  - (c) Morris Worm
  - (d) Code Red
3. To verify a digital signature we need the
  - (a) Sender's Private key
  - (b) Sender's Public key
  - (c) Receiver's Private key
  - (d) Receiver's Public key
4. Which of the following is not a block cipher operating mode?
  - (a) ECB
  - (b) CFB
  - (c) CBF
  - (d) CBC
5. Which of the following characteristics of a programming language is best specified using a context-free grammar?
  - (a) Identifier length
  - (b) Maximum level of nesting
  - (c) Operator precedence
  - (d) Type compatibility

## True / False (10 marks)

*Answer True or False*

6. True or False: The Abstract Syntax Tree (AST) is responsible for managing information about variables and their attributes in a compiler.
7. True or False: A secure pseudorandom permutation (PRP) can be constructed from a secure pseudorandom generator (PRG) using the GGM construction and Luby-Rackoff theorem.
8. True or False: Silk Road was a notorious Dark Web marketplace known for facilitating the sale of illegal drugs and various illicit goods.
9. True or False: Biometric authentication is not considered a secure method for verifying user identity.
10. True or False: The Heartbleed bug enables code injection attacks by overwriting cryptographic keys stored in memory.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Write a function to check whether the given ip address is valid or not using regex.
12. Write a python function to find minimum sum of factors of a given number.

*End of Paper*